

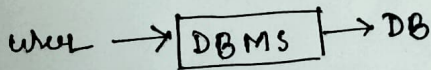
SQL

SQL → Structured Query Language.

Database

Database is collection of data in a format that can be easily accessed (Digital)

A software application used to manage our DB is called DBMS (Data Base Management System)



Types of database

Relational

- Data stored in tables
- Also called RDBMS
- Ex → MySQL
 - II) SQL Server
 - III) ORACLE
 - IV) PostgreSQL

Non-relational

- Data stored in not table format
- Also called NOSQL
- Ex → mongo DB

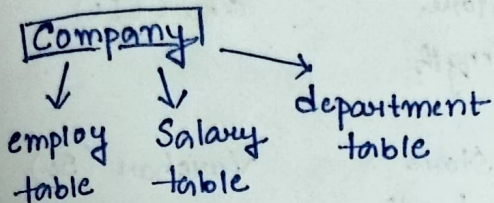
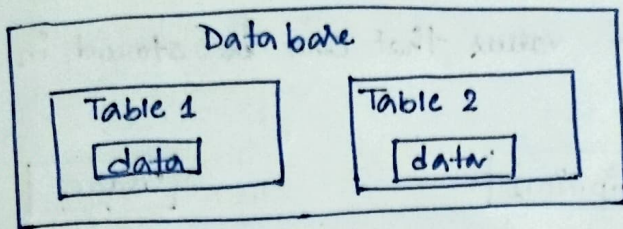
SQL

SQL is a programming language used to interact with relational database.

It is used to perform CRUD operations:

Create
Read
Update
Delete.

Database Structure



Creating database

Create Database db-name; [create new database]

Drop Database db-name; [delete any database]

Creating Table

Use db-name; [To inform that we will go to use this db]

Create table-name (

Column-name 1 datatype constraint,

Column-name 2 datatype constraint,

Column-name 3 datatype constraint,

);

Student.

id	name	age

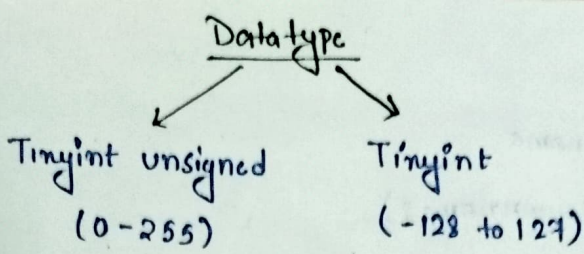


```
Create table Student (
    id int primary key,
    name varchar(50),
    age int not null.
);
```


SQL Datatypes.

They define the type of values that can be stored in a column.

Datatype	Description	Usage
Char	string (0-255), can store characters of fixed length	Char (50)
Varchar	string (0-255), can store characters up to given length	Varchar (50)
Blob	string (0-65535), can store binary large object.	Blob (1000)
Int	integer	int
Tinyint	integer (-128 to 127)	Tinyint
Bigint	integer (-808 to 807)	Bigint
Bit	Can store 2 bit values. 2 can range from 1 to 64	bit (2)
Float	Decimal number with precision fo 23 digits	float
Double	Decimal number with digits 24 to 53	Double
Boolean	Boolean value 0 or 1	Boolean
Date	date in format of YYYY-MM-DD ranging from 1000-01-01 to 9999-12-31	Date
Year	Year in 4 digits format ranging from 1901 to 2155	Year



Types of SQL commands

DDL (Data Definition Language) : create, alter, rename, truncate & drop.

DQL (Data Query Language) : select

DML (Data Manipulation Language) : insert, update, delete

DCL (Data Control Language) : grant & revoke permission to user.

TCL (Transaction Control Language) : start transaction, commit, rollback.

Database related queries.

Create database db-name;

Create database if not exists db-name;

↑ show warning instead of error.

Drop database db-name;

Drop database if exists db-name;

Show databases;

Show Tables;

Table related queries.

(Select and view all columns.)

Select * from table-name;

DQL ↑
All ↑

(Insert data)

Insert into table-name

(columnname1, columnname2)

values

(column1 value1, column2 value1),

(column2 value2, column2 value2);

Insert into Student

(rollno, name)

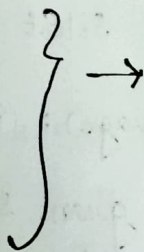
values

(101, "Karan"),

(102, "Arjun");

Student.

roll no	name
101	Karan
102	Arjun



Keys

Primary Key → It is a column (or set of columns) in a table that uniquely identifies each row. (a unique id).

• There is only 1 PK & it should be not null.

Foreign Key → A foreign key is a column (or set of columns) in a table that refers to the Primary Key in different tables.

- There can be multiple foreign keys.
- FK's can have duplicates and null values.

Table 1

(PK) id	name	(FK) city id	city
101	Karan	<u>1</u>	Pune
102	Arjun	2	Mumbai
103	Ram	1	Pune
104	Shyam	3	Delhi

Table 2

(PK) id	cityname
<u>1</u>	Pune
2	Mumbai
3	Delhi

Constraints

SQL constraints are used to specify rules for data in a table.

- (i) Not null columns can't have null values.
- (ii) Unique all values in column can be different.
- (iii) Primary key makes column unique & not null but used only for one.

eg → Primary key (id, name);

id	name
101	Ram
101	Sham
102	Ram

→ Combinations should be different for both columns as Primary key.

- (iv) Foreign key prevents actions that would destroy links between tables.

```
create table temp (
    cust_id int,
    Foreign key (cust_id) references customer(id)
);
```

Another table
↓ column

- (v) Default Sets the default value of a column.

```
Salary int default 25000;
```

↑ It will set this default value if we leave the data blank.

- (vi) Check it can limit the values allowed in a column.

```
create table city (
    id int primary key,
    city varchar(50),
    age int,
    constraints age-check check (age >= 18 and city = "Delhi");
```

```
create table newtab (
    age int check (age >= 18)
```


Select distinct city from table name ;

↑
Non repetitive
value from

↑
this
column

Where clause

To define some conditions.

Select col1, col2 from table-name where condition

Operators in where.

Arithmetic operators → +, -, *, /, %

Comparison operators → =, !=, >, >=, <, <=

Logical operators → AND, OR, NOT, IN, BETWEEN, ALL, LIKE, ANY

Bitwise operators → &, |, ^

Limit clause.

Sets an upper limit on number of (tuples) row to be returned.

Select col1, col2 from table-name Limit number

Order by clause

To sort in ascending or descending order.

Select col1, col2 From table-name order by
'col-name(s)' Asc/Desc ;

Aggregate functions

Aggregate functions performs a calculation on a set of values and return a single value.

- Count ()
- MAX ()
- MIN ()
- SUM ()
- AVG ()

Select max (marks) from students ;

↑ ↑
column Table

Group by clause

Groups rows that have same values into summary rows. It collects data from multiple records and groups the result by one or more column.

Generally we use group by with some aggregation function.

Select city, count(name) from student
group by city;

Having clause

Similar to where, applies some condition on rows. Used when we want to apply any condition after grouping.

Count number of students in each city where max marks cross 90

Select count(name), city
from student.
group by city.

Having max(marks) > 90;

General order

Select column(s)
from table-name
where condition → Apply on rows
group by column(s)
having condition → Apply on groups
order by column(s) ASC;

Table related queries

• Update → to update existing rows.

update table-name
set col1 = val1, col2 = val2
where condition;

{ update student
set grade = "O"
where grade = "A";

- Delete → To delete existing rows

Delete from table_name
where condition;

{ Delete from student
where marks > 33;

Cascading in foreign key

On delete cascade → when we create a foreign key using this option it deletes the referencing rows in the child table when the referenced row is deleted in the parent table which has a primary key.

On update cascade → when we create a foreign key using update cascade the referencing rows are updated in the child table when the referenced row is updated in the parent table which has a primary key.

Create table Student (
id int primary key,
courseid int

Foreign key (courseid) references course (id)

on delete cascade
on update cascade
);

Table related query

- Alter → to change the schema.

(To add a column) → Alter table table_name
Add column column_name data type constraint;

(To drop a column) → Alter table table_name
Drop column column_name.

(To rename table) → Alter table table_name
Rename to new_table_name;

(To rename column name) → Alter table table-name
Change column old-name
new-name new-datatype new-constraint

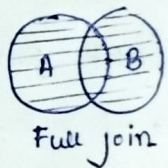
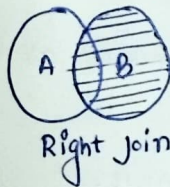
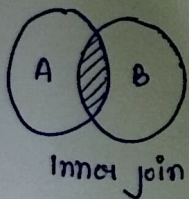
(To modify column ~~name~~) → Alter table table-name
Modify col-name new-datatype
new-constraint;

• Truncate → To delete table's data.

Truncate Table table-name;

Joins in SQL

Join is used to combine rows from two or more tables, based on a related column between them.



Inner join → Return records that have matching values in both tables.

```
Select column(s)
from table A
Inner join table B
on table A.col-name = table B.col-name;
```

Left join → Returns all records from the left side table and the matched records from the right table.

```
Select column(s)
from table A
Left join table B
on table A.col-name = table B.col-name;
```


Full join

Return all records when there is a match in either left or right table.

Select * from student as a

Left join course as b

on a.id = b.id

union

Select * from student as a

Right join course as b

on a.id = b.id.

Left/Right exclusive join

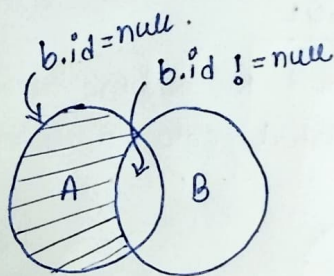
Select *

from student as a

Left/Right join course as b

on a.id = b.id

where b.id is null.



Left exclusive

Self join

It is a regular join but the table is joined with itself.

Select column(s)

from table as a

join table as b

on a.col-name = b.col-name;

Union

It is used to combine the result set of two or more Select Statements. Gives unique records.

To use it →

- Every Select should have same no of columns.
- Columns must have similar data types.
- Columns in every Select should be in same order.

SQL Sub Queries.

A subquery or inner query or a nested query is a query within another SQL query.

It involves 2 select statements.

```
Select column(s)
from table-name
where col-name operator
(subquery);
```

```
{ Select employeeid, name, salary
  from employee
  where salary >
    ( Select AVG(salary) from
      employee );
```

MySQL Views

A view is a virtual table based on the result-set of an SQL statement.

```
Create view view1 as
```

```
Select rollno, name from student;
```

```
Select * from view1;
```

* A view always shows up to date data. The database engine recreates the view, every time a user queries it.